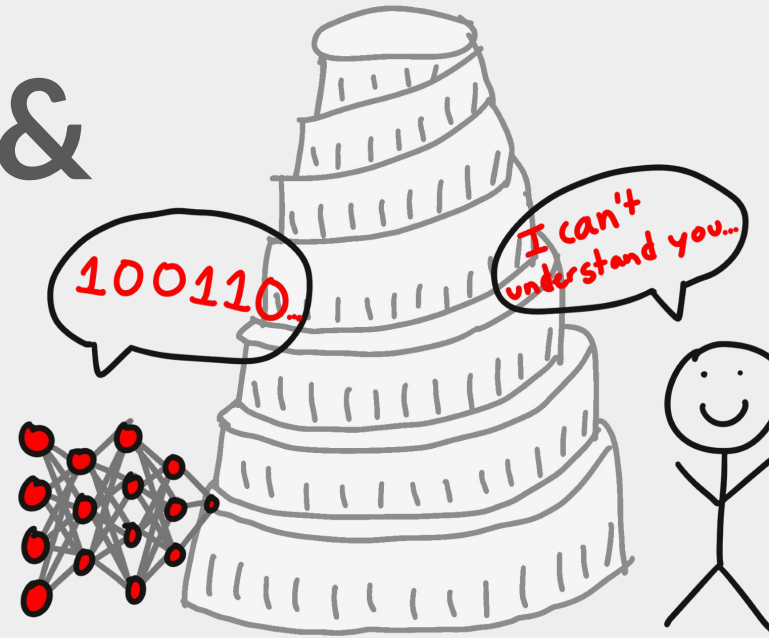


Interpreting “Interpretability” & Explaining “Explainability” In ML and Physics



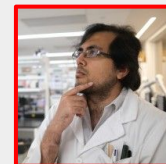
Pictured: Genesis of Uninterpretability

Rikab Gambhir

gambhirb@ucmail.uc.edu

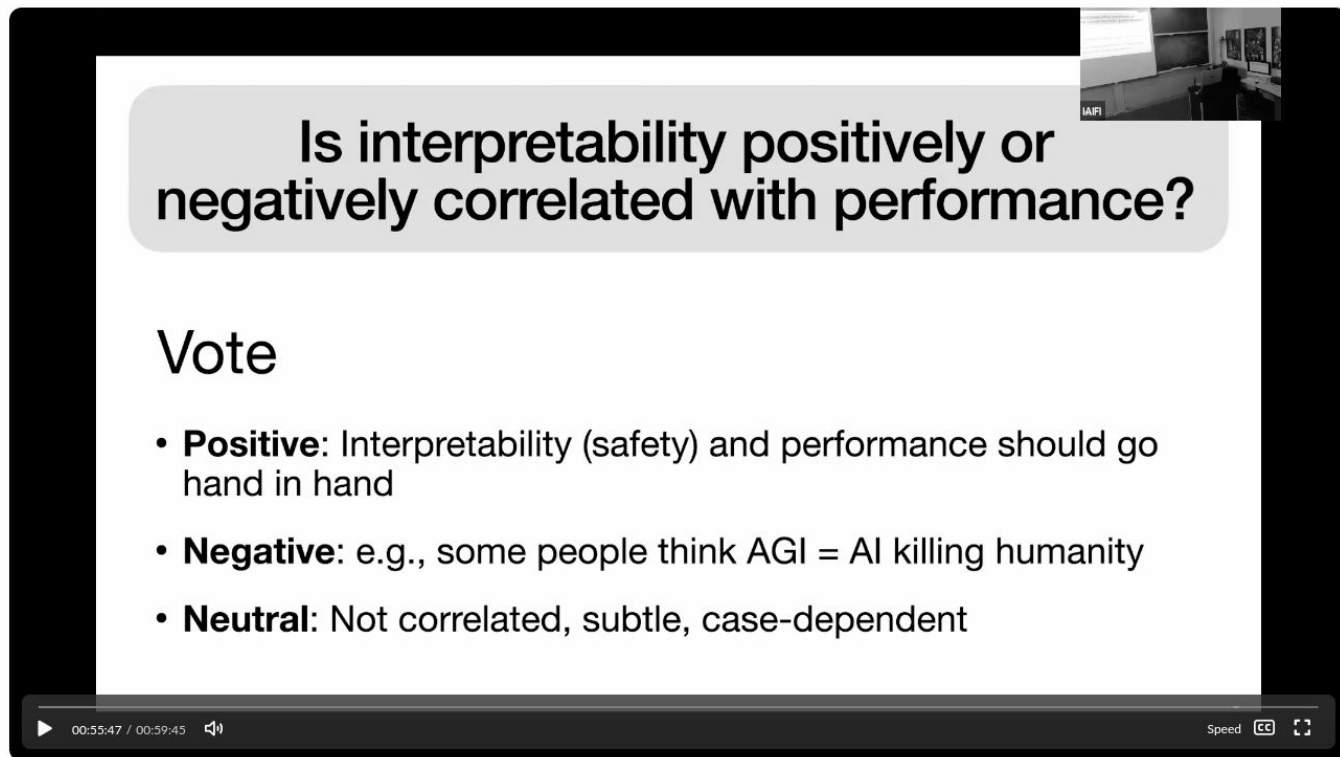
In Collaboration with Luisa Lucie-Smith and Jesse Thaler

IAIFI Lunch Discussion



On April 18, 2025: A **Legendary IAIFI Event**

IAIFI Thematic Discussion on interpretability in ML



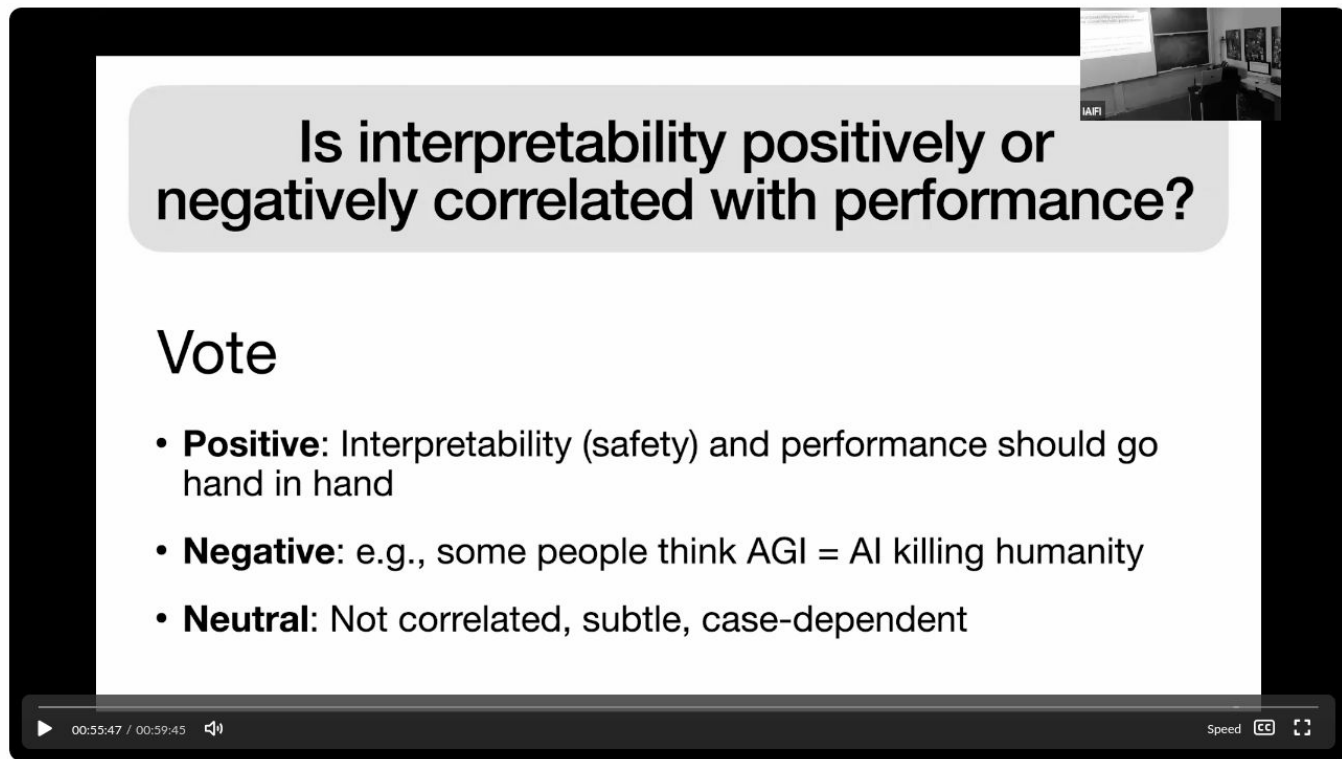
Is interpretability positively or negatively correlated with performance?

Vote

- **Positive:** Interpretability (safety) and performance should go hand in hand
- **Negative:** e.g., some people think AGI = AI killing humanity
- **Neutral:** Not correlated, subtle, case-dependent

On April 18, 2025: A **Legendary IAIFI Event**

IAIFI Thematic Discussion on interpretability in ML



Is interpretability positively or negatively correlated with performance?

Vote

- **Positive:** Interpretability (safety) and performance should go hand in hand
- **Negative:** e.g., some people think AGI = AI killing humanity
- **Neutral:** Not correlated, subtle, case-dependent

... I remember being extremely confused by this question!

On April 18, 2025: A **Legendary IAIFI Event**

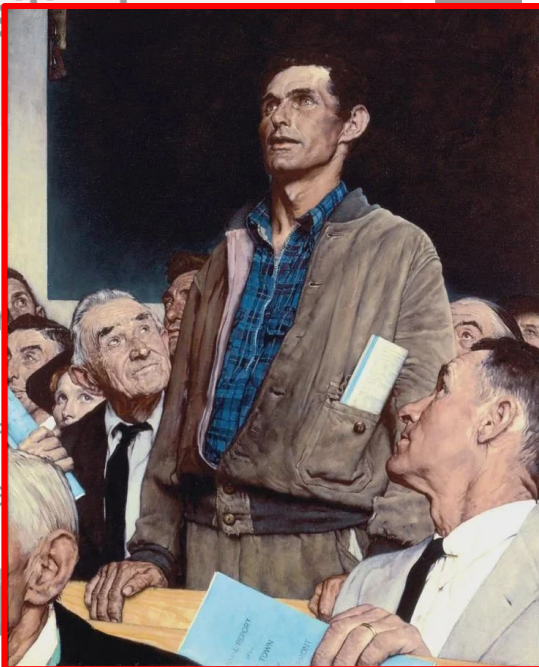
IAIFI Thematic Discussion on interpretability in ML

Is interpretability positively or negatively correlated with performance?

Vote

- **Positive:** Interpretability (safety) and performance go hand in hand
- **Negative:** e.g., some people think AGI = interpretability
- **Neutral:** Not correlated, subtle, case-dependent

00:55:47 / 00:59:45



Audio Transcript

Q Search transcript

right now, but I guess this is where we are working.

Can't define performance without some notion of interpretability. You can be, you know, in jet physics we we train increasingly better classifiers on our simulations, and they get better and better and better every year.

It doesn't actually mean anything that these, that they, your classifier score is better. If you can't interpret the score is, it's not really useful for anything. If you're limited by like, how good your simulation is, or how good your data is, or you can't do anything with this data.

You're you have to build it interpreted into your metric performance in the 1st place.

Otherwise, you don't know that you've done better.

So I mean, you're basically stating the position that interpretability is about the outcome, and that the thing that we want, at least in this, in the scientific machine learning context is that the outcome actually is the thing that we asked for. Yes, and if the outcome is the thing we asked for, and more interpretable means more aligned with actually what scientifically we defined the problems to be.

↑ Resume autoscrolling

... I remember being extremely confused by this question!

Outline

My goal today is to give you an overview of my (*disclaimer: potentially biased*) view of interpretability and explainability in ML and physics, framed around these 3 questions:

1. What is (and isn't) **interpretability** and **explainability**?
2. Why (and why not) **interpretability** and/or **explainability**?
3. Do we even need **interpretability** and/or **explainability**?

Outline

My goal today is to give you an overview of my (*disclaimer: potentially biased*) view of interpretability and explainability in ML and physics, framed around these 3 questions:

1. What is (and isn't) **interpretability** and **explainability**?

*Our ability to understand, or approximate
what a model is.*

*Our ability to relate a model to prior domain
knowledge.*

2. Why (and why not) **interpretability** and/or **explainability**?

3. Do we even need **interpretability** and/or **explainability**?

Outline

My goal today is to give you an overview of my (*disclaimer: potentially biased*) view of interpretability and explainability in ML and physics, framed around these 3 questions:

1. What is (and isn't) **interpretability** and **explainability**?

Our ability to understand, or approximate what a model is.

Our ability to relate a model to prior domain knowledge.

2. Why (and why not) **interpretability** and/or **explainability**?

Trustworthiness, uncertainty, and model expressivity.

Knowledge extraction, validation, and data adaptability.

3. Do we even need **interpretability** and/or **explainability**?

Outline

My goal today is to give you an overview of my (*disclaimer: potentially biased*) view of interpretability and explainability in ML and physics, framed around these 3 questions:

1. What is (and isn't) **interpretability** and **explainability**?

Our ability to understand, or approximate what a model is.

Our ability to relate a model to prior domain knowledge.

2. Why (and why not) **interpretability** and/or **explainability**?

Trustworthiness, uncertainty, and model expressivity.

Knowledge extraction, validation, and data adaptability.

3. Do we even need **interpretability** and/or **explainability**?

I don't know – you tell me!

(Task specification, end goals, model design, and intervention)

Sometimes, interpretability is performance!



~Yoda, on ML Models

First: ML Models vs. “Classical” Physics Models

A fundamental axiom we will adopt is that ML Models are **no different** than “ordinary” (classical*) physical models, except in size and scope.

All models (the Standard Model, Λ CDM, Newton’s Law of Gravitation, or any ML model you come up with) are merely parameterized functions with fits to data. ML models just typically have more parameters.

Why do physics-based models *feel* nicer? **Interpretability** and **Explainability**, defined shortly!

*By classical, I do not mean non-quantum or non-relativistic. Just models that a pre-ML physicist would derive.

Some Definitions

I will simply assert the following, whether or not these definitions are good is worth discussing!

Interpretability: The ability to understand, reduce, or approximate a model *mechanistically*.

“Do I, as a human, understand what the model even is?”

Explainability: The ability to associate the model, or its subcomponents, to some prior *domain knowledge*.

“How do I extract scientific value from my model?”

In this talk, I will associate *interpretability* with *red* and *explainability* with *blue*.

The primary color of the talk is also red, to remind you that interpretability usually precedes explainability

Some Definitions

I will simply assert the following, whether or not these definitions are good is worth discussing!

Interpretability: The ability to understand, reduce, or approximate a model *mechanistically*.

“Do I, as a human, understand what the model even is?”

Context Independent: Intrinsic to the model definition

Explainability: The ability to associate the model, or its subcomponents, to some prior *domain knowledge*.

“How do I extract scientific value from my model?”

Context Dependent: Relates to data, other models, and/or a specific science task

In this talk, I will associate *interpretability* with *red* and *explainability* with *blue*.

The primary color of the talk is also red, to remind you that interpretability usually precedes explainability

*Interpretability is about a user's ability to understand, and is therefore subjective. In this talk, "interpretable" will really mean "interpretable to Rikab", which hopefully correlates with "interpretable to you"

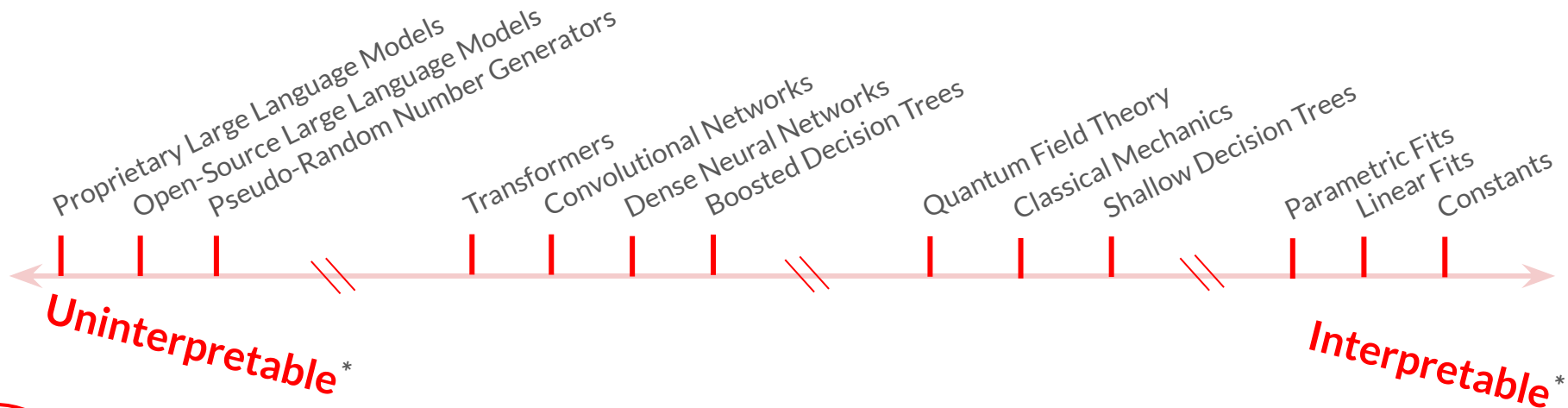
Some Examples

Interpretability: The ability to understand, reduce, or approximate a model *mechanicalistically*.

"Do I, as a human, understand what the model even is?"

Models I *don't* know how to write down, compute, or approximate

Models I know how to write down, compute, or approximate



*Interpretability is about a user's ability to understand, and is therefore subjective. In this talk, "interpretable" will really mean "interpretable to Rikab", which hopefully correlates with "interpretable to you"

Some Examples

Interpretability: The ability to understand, reduce, or approximate a model *mechanistically*.

"Do I, as a human, understand what the model even is?"

I like to think of interpretability in terms of information *about* a model, emotionally similar to a description length or **Kolmogorov Complexity**

All of these increase our information *about* a model:

- Metrics, including losses (This is why I was confused earlier! Performance is a type of interpretability.)
- Approximations
- Mechanistic code / instructions
- Simple mathematics

Interpretability is what separates "classical" models from "ML" ones!

*Explainability does not exist outside the context of a scientific problem with domain knowledge to relate to. In this talk, “explainable” will really mean “in relation to collider physics and related problems Rikab works on”, and you should extrapolate to your field.

Some Examples

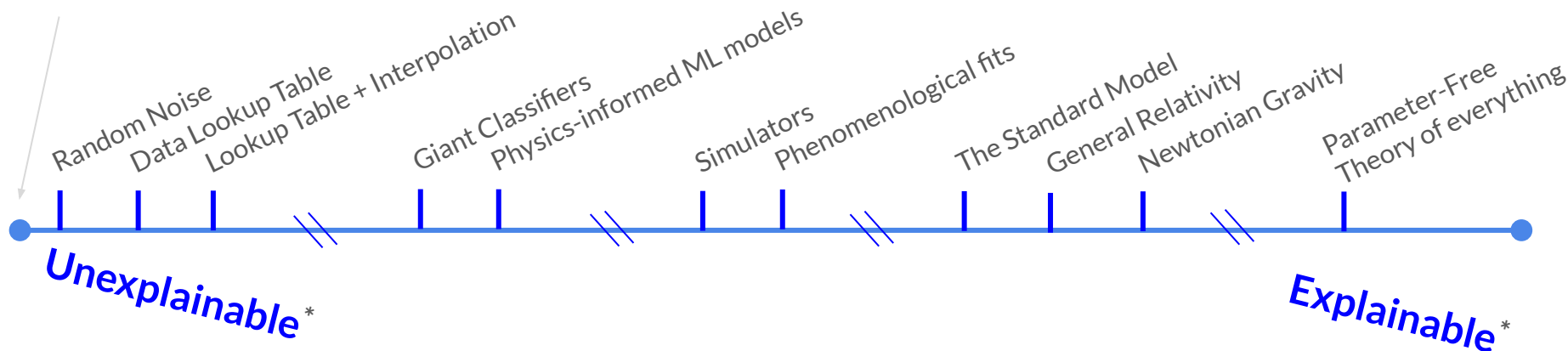
Explainability: The ability to associate the model, or its subcomponents, to some prior **domain knowledge**.

“How do I extract scientific value from my model?”

Models I can't relate to physics I already know

Models I *can* relate to physics I already know

Why an endpoint and not an arrow?
Ask me after!



Some Examples

Explainability: The ability to associate the model, or its subcomponents, to some prior **domain knowledge**.

“How do I extract scientific value from my model?”

Explainability does not exist without a prior model or domain knowledge *to compare against*.

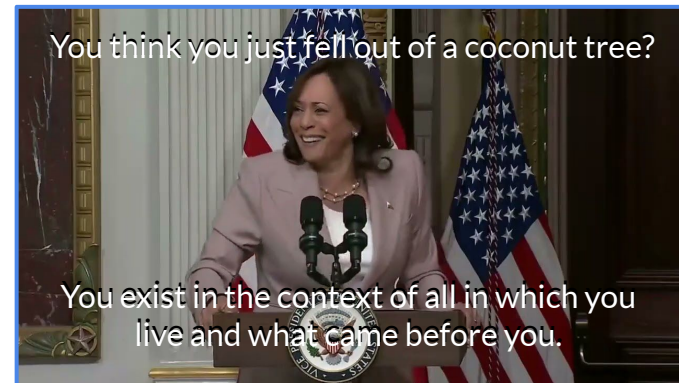
Knowledge is inherently relational!

The Yoneda Lemma

For any functor $F: \mathcal{C}^{op} \rightarrow \mathbf{Set}$ and any object X in $\mathcal{C}$, natural transformations $\text{hom}(-, X) \rightarrow F$ are in bijection with elements in the set $F(X)$. That is,

$$\text{Nat}(\text{hom}(-, X), F) \cong F(X).$$

Same Idea!



The Landscape

“Surrogate Models”

- Differential Likelihood Estimators
- Simulators / generative models
- Numeric Approximators

Explainable

“Classical Models”

- First-Principles Physics Models
- Shallow Decision Trees
- Qualitative Models

These axes
aren't really
orthogonal

Uninterpretable

Interpretable

“Artificial Intelligence”

- Proprietary Black Boxes
- LLMs
- Agentic AI

Unexplainable

“Data Fits”

- Lookup Tables
- ML Taggers / Regressors
- High Dimensional Linear Regressions
- “Second-Principles” phenomenological models (e.g. hadronization)

To Ponder! Are these models *science*, or are these just tools?

A worked example

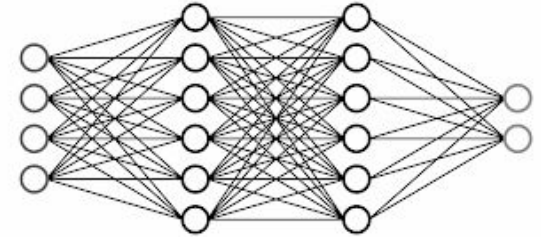
Suppose we want to predict the motion of heavenly bodies given their masses and positions, based on astronomical data. Let's devise two models.

Model A



*Shamelessly generated with Gemini because I can't draw faces that well. Sorry!

A model with many free weights and biases, fit to astronomical data.



Abstractly, a neural network to calculate forces given positions and masses

Model B



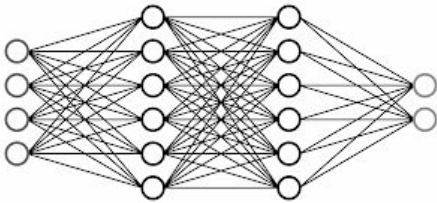
A model with a single free parameter G , fit to astronomical data.

$$F = G \frac{m_1 m_2}{r^2}$$

Abstractly, a simple function to calculate forces given positions and masses

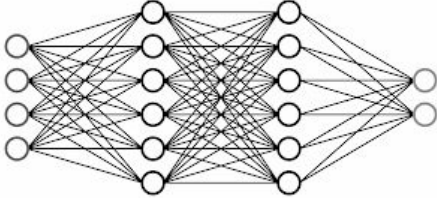
How are these models different?

Fundamentally, they aren't. It's just a matter of degree and tradeoffs.

	Model A	Model B
Computation	 Difficult	$F = G \frac{m_1 m_2}{r^2}$ Easy
Parameters	Many Parameters	One Parameter G
Accuracy	Higher	Lower
Precision	Lower	Higher

How are these models different?

Fundamentally, they aren't. It's just a matter of degree and tradeoffs.

	Model A	Model B
Computation	 Difficult	$F = G \frac{m_1 m_2}{r^2}$ Easy
Parameters	Many Parameters	One Parameter G

Semantically Meaningful Parameters

It is not obvious how any parameter in Model A affects the output force of gravity. It is obvious how G in Model B affects the output force of gravity: proportionally! Therefore, G derives semantic meaning: "Strength of Gravity"

Semantic Meaning of Parameters = Shortcuts to calculation

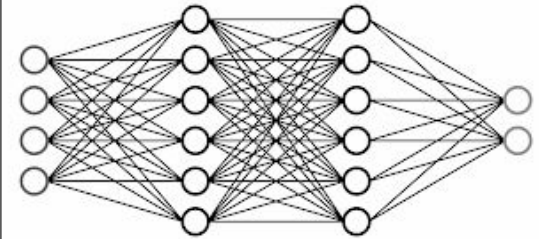
Interpretability: A worked example

I want to *interpret* these models — understand these functions and their properties beyond just plugging in numbers.

- In model B, I can *guarantee* that closed orbits form. In model A, I cannot, since I can barely even write the functional form.
- In model B, I can tell you exactly what happens when masses or distances double, without having to actually compute anything. In model A, I cannot.
- In model B, I can tell you precisely what happens if the parameter G is changed. In model A, I cannot.

In the simpler model B, it is easier to reason about and check properties of the function. It is *more interpretable* and we can assign *semantic meaning*.

Model A



Abstractly, a neural network to calculate forces given positions and masses

Model B

$$F = G \frac{m_1 m_2}{r^2}$$

Abstractly, a simple function to calculate forces given positions and masses

“So what if my model is interpretable?”

An interpretable model is more **trustable**, and can become **explainable**.

Robustness

Explainability

Transparency

Guarantees

Validity / Uncertainty

“So what if my model is interpretable?”

An interpretable model is more **trustable**, and can become **explainable**.

Robustness

“Is it OK if my parameters are slightly wrong?”

Explainability

“Can I explicitly compare to other models?”

Transparency

“How will I tell you what my model is? Does it work as advertised?”

Guarantees

“Are there useful properties that are always true?”

→ “Are there *physical* properties that are always true?”

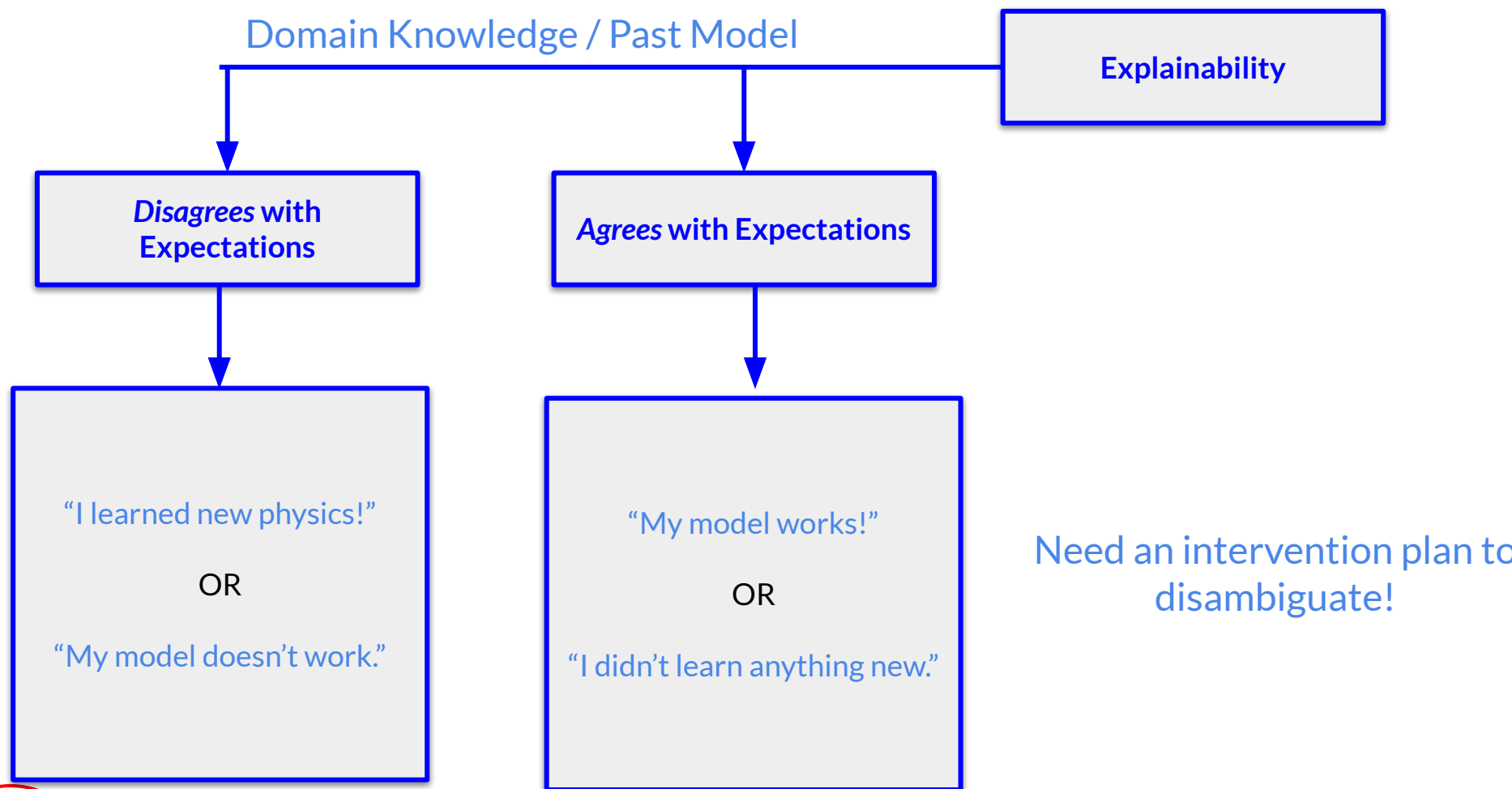
Validity / Uncertainty

“Can I quantify model or numeric uncertainties or domains of validity?”

These questions require interpretability to answer!

“So what if my model is explainable?”

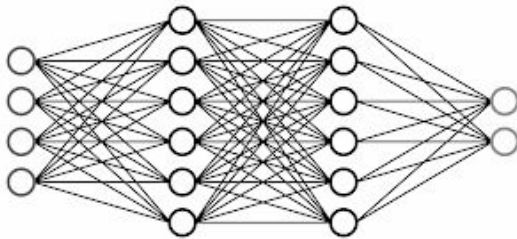
An explainable model can be **verified** or ^{*} used for **scientific discovery**



Post-Hoc Tools

Model *B* is *intrinsically* interpretable, by design. It was constructed with only interpretable parts. Model *A* isn't. That doesn't mean we can't try to interpret it after the fact!

Model A



Often, this is done to achieve some amount of **explainability**.

Once a non-interpretable model is interpreted post-hoc, one can compare it to other existing scientific models to see if it either aligns with expectations, or more interestingly, if it doesn't.



I can't interpret large piles of weights and biases.

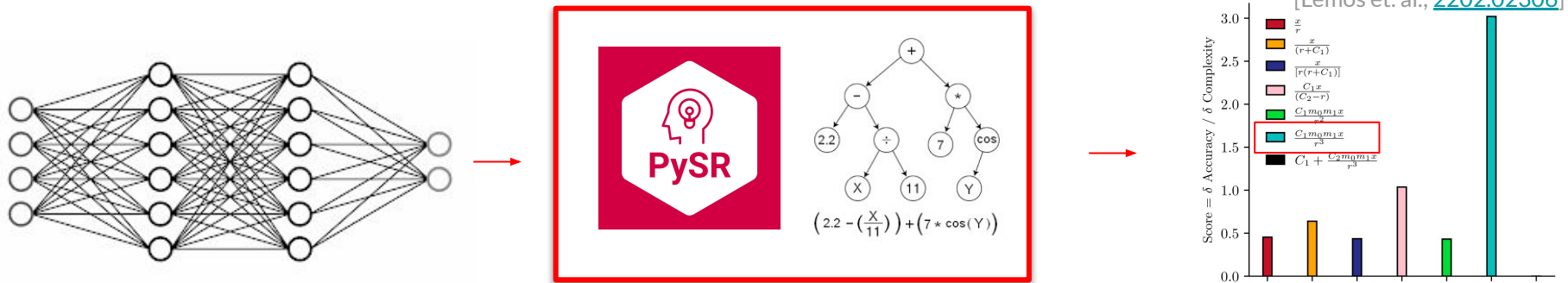
But can I get *anything* out of this?

Is it similar to my prior Model B that I already know?

Post-Hoc Tools: Knowledge Distillation

Knowledge distillation is the art of training an intrinsically interpretable model to mimic a less interpretable model. A very common example is **symbolic regression**, which finds a symbolic approximation by minimizing some complexity measure.

Model A



PySR is a commonly used package for symbolic regression



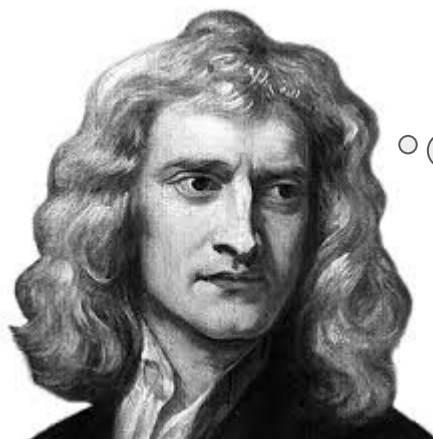
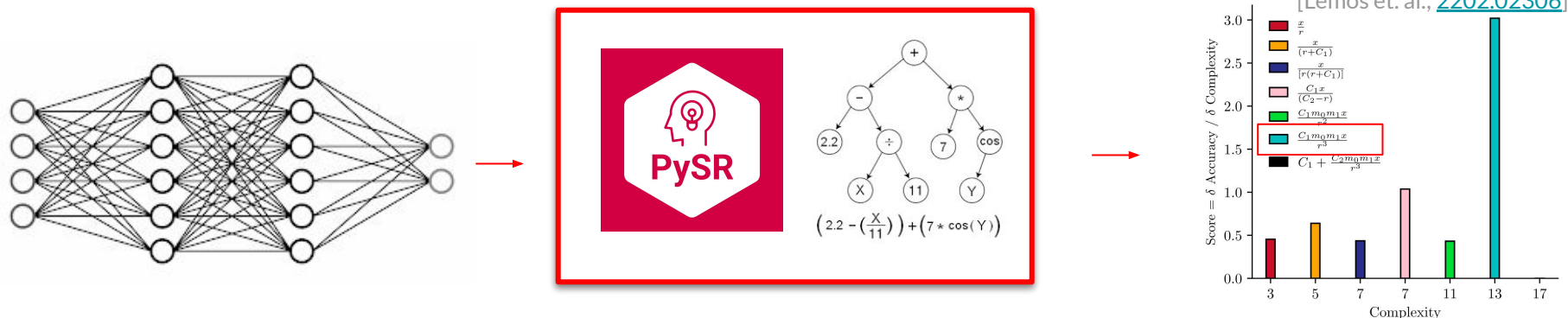
$$F = G \frac{m_1 m_2}{r^2}$$

[Cranmer; [2305.01582](#)]
[Cranmer et. al.; [2006.11287](#)]

Post-Hoc Tools: Knowledge Distillation

Knowledge distillation is the art of training an intrinsically interpretable model to mimic a less interpretable model. A very common example is **symbolic regression**, which finds a symbolic approximation by minimizing some complexity measure.

Model A



I can approximate Model A as a much simpler model that I can interpret. Now I can partially interpret Model A!

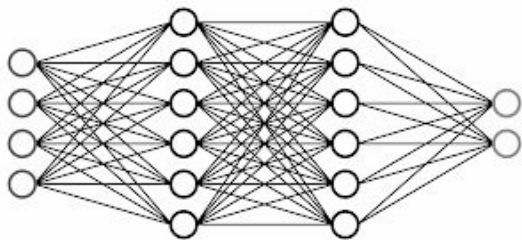
$$F = G \frac{m_1 m_2}{r^2}$$

[Cranmer; [2305.01582](#)]
[Cranmer et. al.; [2006.11287](#)]

Post-Hoc Tools: Feature Metrics

Even if we don't have a closed form for our model $f(x)$, we can still roughly estimate how much each input feature x_i contributes to the output. A common metric is **SHAPLEY values**.

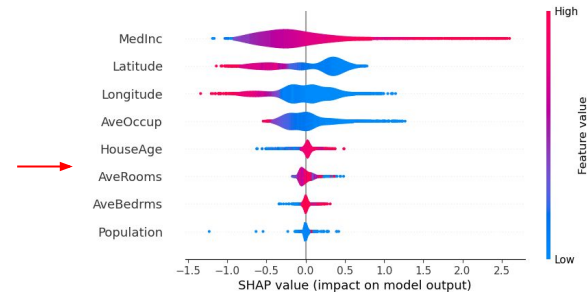
Model A





$$\begin{aligned}\varphi_i(v) &= \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} (v(S \cup \{i\}) - v(S)) \\ &= \frac{1}{n} \sum_{S \subseteq N \setminus \{i\}} \binom{n-1}{|S|}^{-1} (v(S \cup \{i\}) - v(S))\end{aligned}$$

SHAP is a package for estimating SHAPLEY values



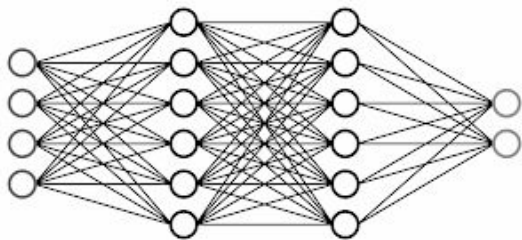
Roughly, Average contribution of each input to the final output



Post-Hoc Tools: Feature Metrics

Even if we don't have a closed form for our model $f(x)$, we can still roughly estimate how much each input feature x_i contributes to the output. A common metric is **SHAPLEY values**.

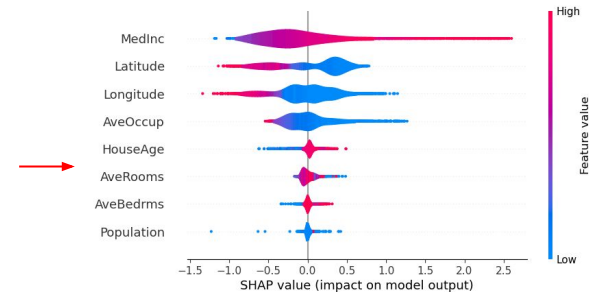
Model A



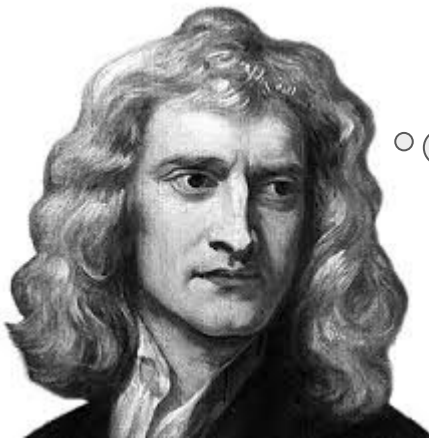


$$\begin{aligned}\varphi_i(v) &= \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (n - |S| - 1)!}{n!} (v(S \cup \{i\}) - v(S)) \\ &= \frac{1}{n} \sum_{S \subseteq N \setminus \{i\}} \binom{n-1}{|S|}^{-1} (v(S \cup \{i\}) - v(S))\end{aligned}$$

SHAP is a package for estimating SHAPLEY values



Roughly, Average contribution of each input to the final output

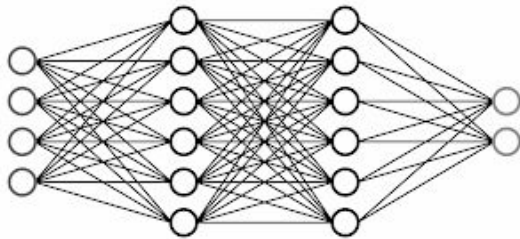


Now I can tell that more massive objects contribute more to force! I understand Model A a little better.

2. Why (and why not) interpretability?

Post-Hoc Interpretability: **Caution! Case I**

Once a model is interpreted, one must be careful how they will relate it back to prior domain knowledge (**explainability**). If it matches your expectations, will you increase your trust in the model? If it doesn't, do you decrease or trust in the model, or accept your prior model was wrong?
One needs an "intervention" plan, decided before you see the data.



Knowledge Distillation



$$V = -G \frac{Mm}{r} + \frac{L^2}{2\mu r^2}$$

The model learned exactly what I expected it to!
I trust Model A more now.

OR

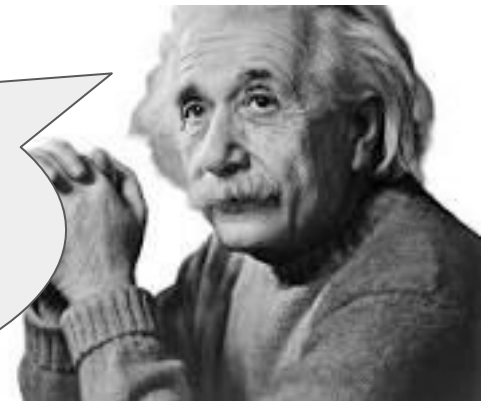
Model A is useless! I already knew this!



The model failed to learn what I expected! Model A or the symbolic regression must be wrong, I no longer trust Model A.

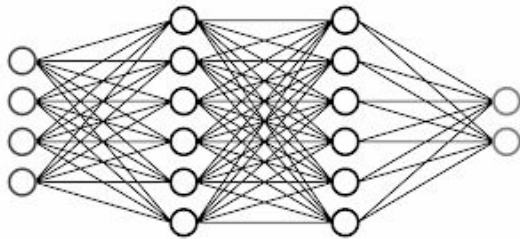
OR

Model A learned something I didn't expect!
Perhaps it was my previous understanding that was wrong.



Post-Hoc Interpretability: **Caution! Case II**

Once a model is interpreted, one must be careful how they will relate it back to prior domain knowledge (**explainability**). If it matches your expectations, will you increase your trust in the model? If it doesn't, do you decrease or trust in the model, or accept your prior model was wrong?
One needs an "intervention" plan, decided before you see the data.



Knowledge Distillation

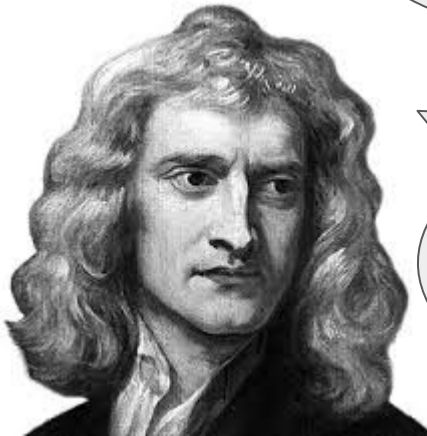


$$V = -G \frac{Mm}{r} + \frac{L^2}{2\mu r^2} + G \frac{(m + M)L^2}{c^2 \mu r^3}$$

The model learned exactly what I expected it to!
I trust Model A more now.

OR

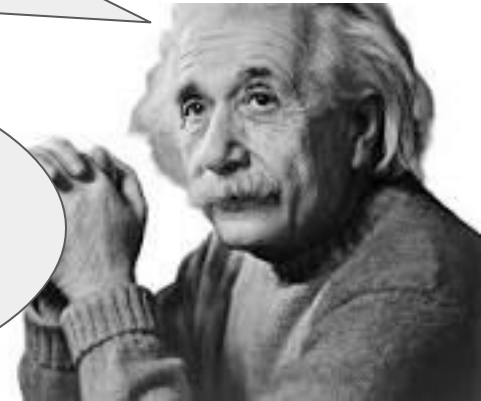
Model A is useless! I already knew this!



The model failed to learn what I expected! Model A or the symbolic regression must be wrong, I no longer trust Model A.

OR

Model A learned something I didn't expect!
Perhaps it was my previous understanding that was wrong.



Post-Hoc Interpretability: **Caution! Case II**

Once a model is interpreted, one must be careful how they will relate it back to prior domain knowledge (**explainability**). If it matches your expectations, will you increase your trust in the model? If it doesn't, do you decrease or trust in the model, or accept your prior model was wrong? *One needs an "intervention" plan, decided before you see the data.*

Key Takeaway: Once you have interpreted a model, what will you actually do with that interpretation?

What will you do if it aligns with what you expected?

What will you do if it *doesn't* align with what you expected?

Or, did you even need the interpretation at all?

Be careful of confirmation bias! Plan ahead, the decision to interpret and your actions thereafter is itself part of the model design.



OR
Model A learned something I didn't expect!
Perhaps it was my previous understanding that
was wrong.



2. Why (and why not) interpretability?

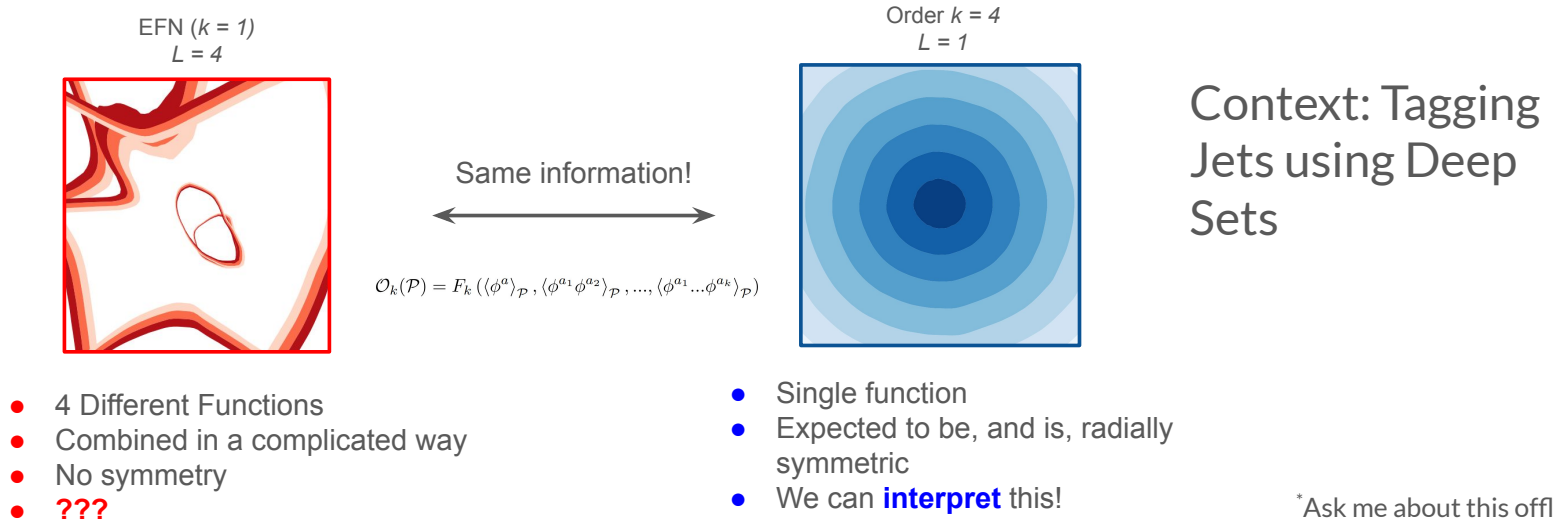
Intrinsic Interpretability

Models are a choice! *Including the number of latent space dimensions!*

My mortal brain can only interpret simple 1D functions, not many 2D functions at once.

For me to gain interpretability, the choice is clear, only use 1 function! But I don't want to lose information...

We can use a trick called **Moment Pooling**^{*} to compress Deep Sets information into fewer dimensions without loss of performance



Interpretability → Explainability

[Komiske, Metodiev, Thaler; [1810.05165](#)]
[RG, Osathapan, Thaler; [2403.08854](#)]

We can extract physics from this!

$$\Phi_{\mathcal{L}}(r) = c_1 + c_2 \log(c_3 + r)$$

This defines a **log angularity** observable, related to the $\square \rightarrow 0$ limit of ordinary angularities

We can even see non-perturbative physics!

$$c_3 \sim \frac{\Lambda_{\text{QCD}}}{p_T R} \sim 0.001$$

Just this alone is enough to get an IRC-safe quark-gluon classifier with an AUC ~ 0.83 !

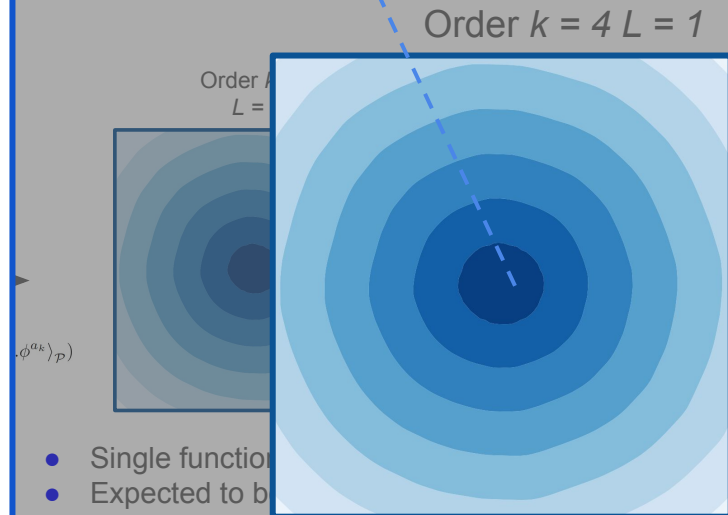
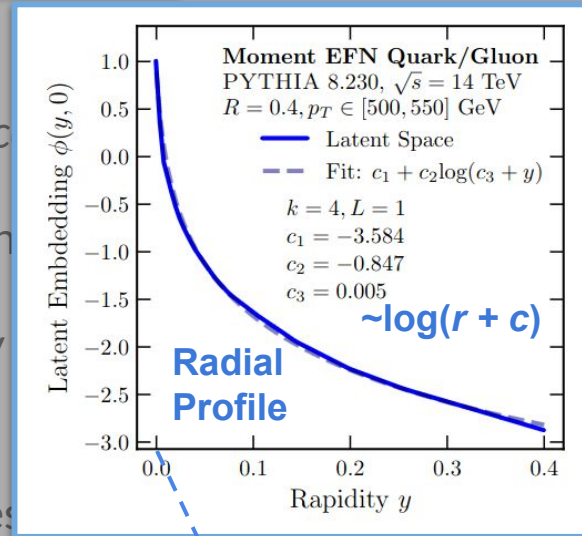
- No symmetry
- ???

tent space

function

ear, only

compress



- Single function
- Expected to be symmetric
- We can **interpret** this!

*Ask me about this offline!

Another Example: Building in the Structure You Want to Interpret!

I have a problem where I want to fit probability distributions, but then interpret those distributions later as perturbative expansions.

Normalizing flows don't cut it: They can fit probability distributions, but I can't read a perturbative expansion off of them.

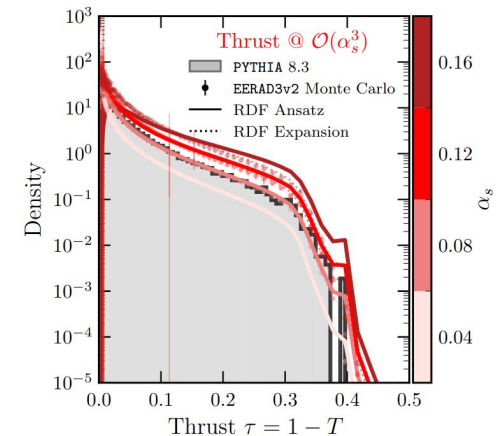
Solution: Build a distribution approximator (we call it the **Resummed Distribution Function**) with perturbative expansions made explicit!

$$q(t|\alpha) = g^*(t, \alpha) \exp \left[-g_{\text{Analytic}}(t, \alpha) - \int_0^t dt' g^*(t', \alpha) \exp(-g_{\text{Analytic}}(t', \alpha)) \right]$$

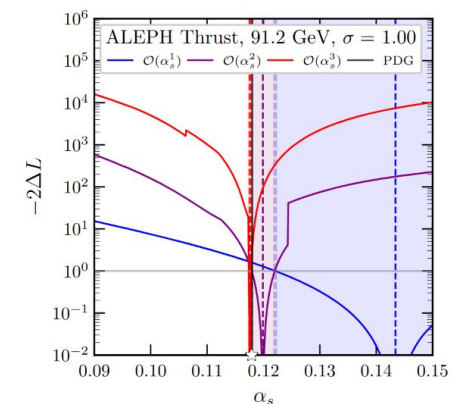
$$g^*(t, \alpha), g_{\text{Analytic}}(t, \alpha) = \sum_{m=0, n=0}^{M, N} g_{mn} \alpha^m t^n \Theta(t - \theta_{mn})$$

Emotionally, this ansatz is like an energy-based model but we made it Taylor-expandable explicitly.

Built-in perturbative coefficients I can read off. Here, interpretability = performance!



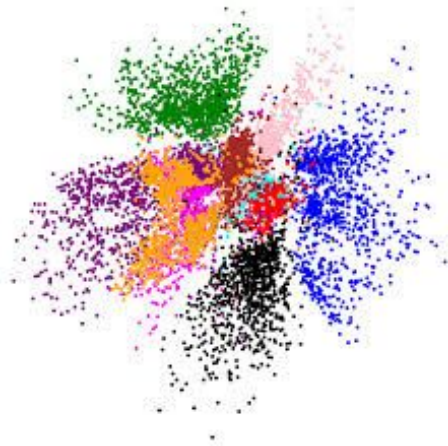
I can now fit and match and interpret my result *exactly* as a QCD expansion!



An Annoying Technicality: Explaining Tea Leaves

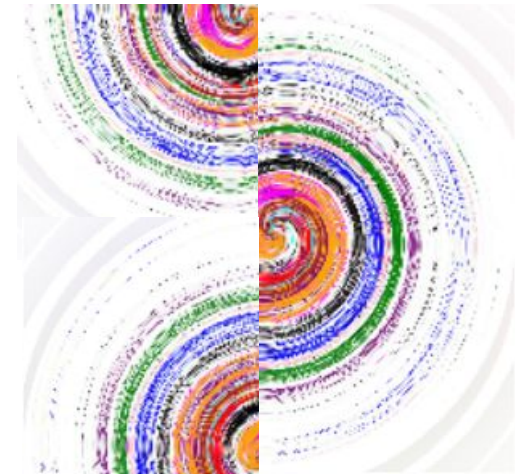
Be careful when looking too deeply at the structure of model latent spaces!

Networks are almost-universal-function approximators. That means that representations have almost-infinite diffeomorphism invariance, and the only thing selecting a representation is the finiteness of the network.



An Autoencoder MNIST Latent Space

Same information!
↔



A simple coordinate transformation

*Feel free to email me afterwards to ask about my Autoencoder rant!

An Annoying Technicality: Explaining Tea Leaves

The only way to break this degeneracy is by:

- Explicitly imposing some structure or regulator based on your domain knowledge
- Letting the whims of gradient descent and finite scaling laws decide for you.



Narcissus, staring into a latent space

But be careful! Often, the structure of a latent space is just a reflection of the inductive bias you put in, either deliberately or accidentally.

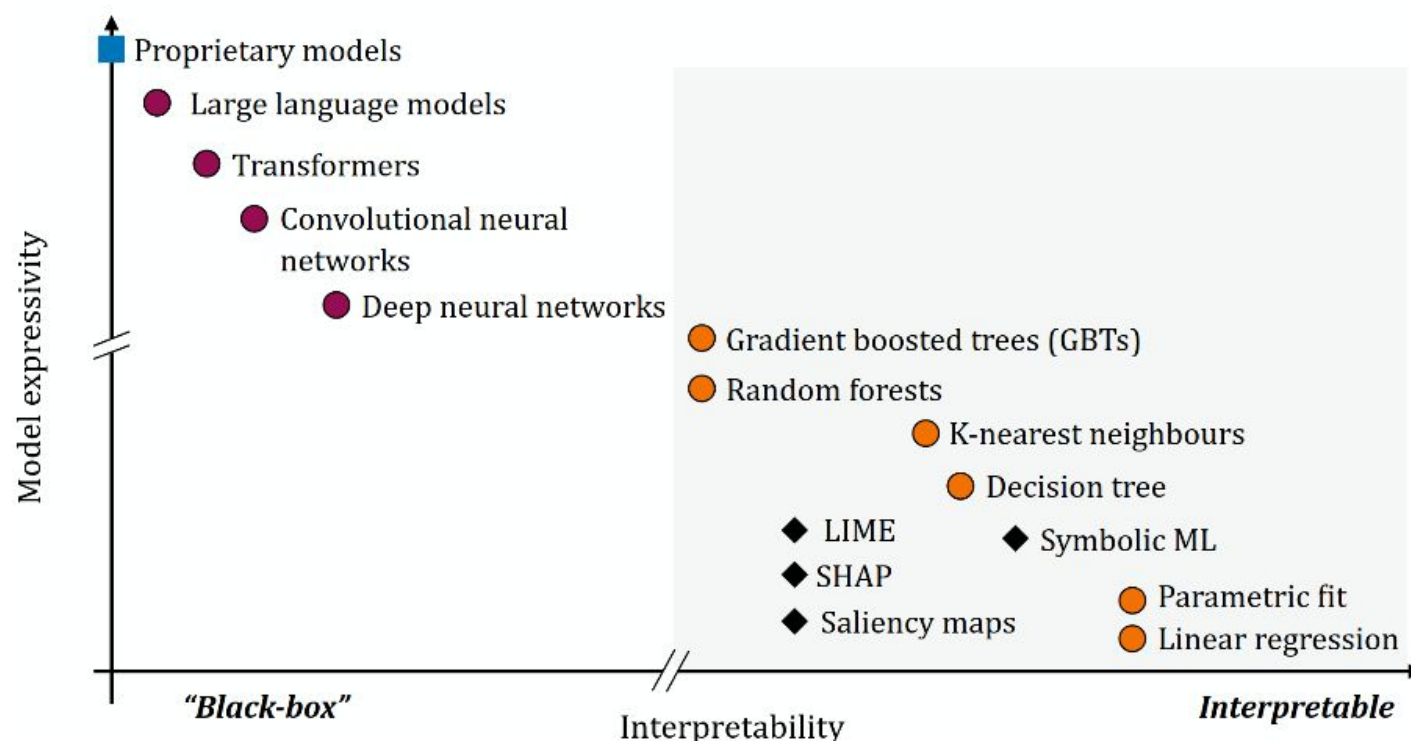
Unless you have some specific reason to expect your structure will provide a desirable solution, you usually don't gain any information you didn't already have.

Watch out for confirmation bias! Don't get obsessed with your own reflection.

Ask me later for a fun coincidental story about this painting!

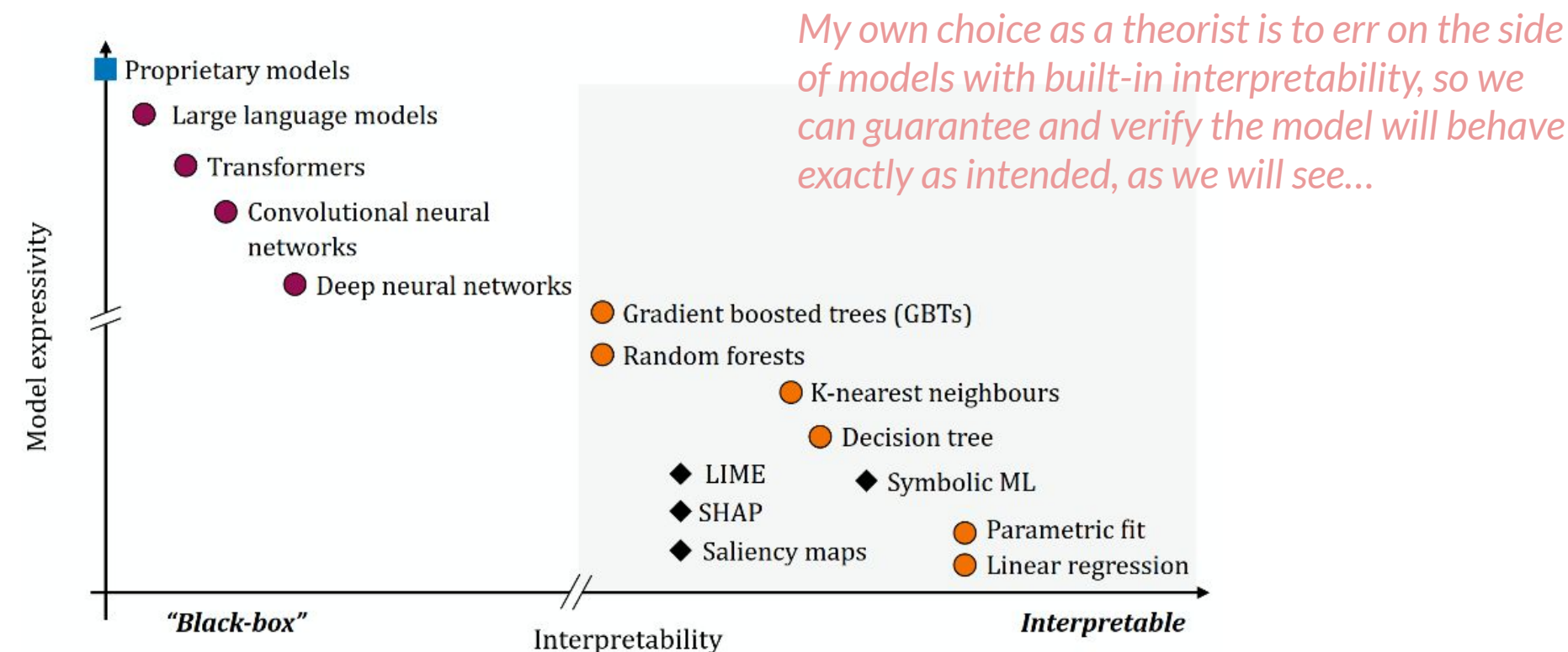
Interpretability is a choice.

Model *B* has built-in interpretability. Model *A* does not. That does not mean we should throw out Model *A*! **Interpretability vs. Expressivity is a trade off.**



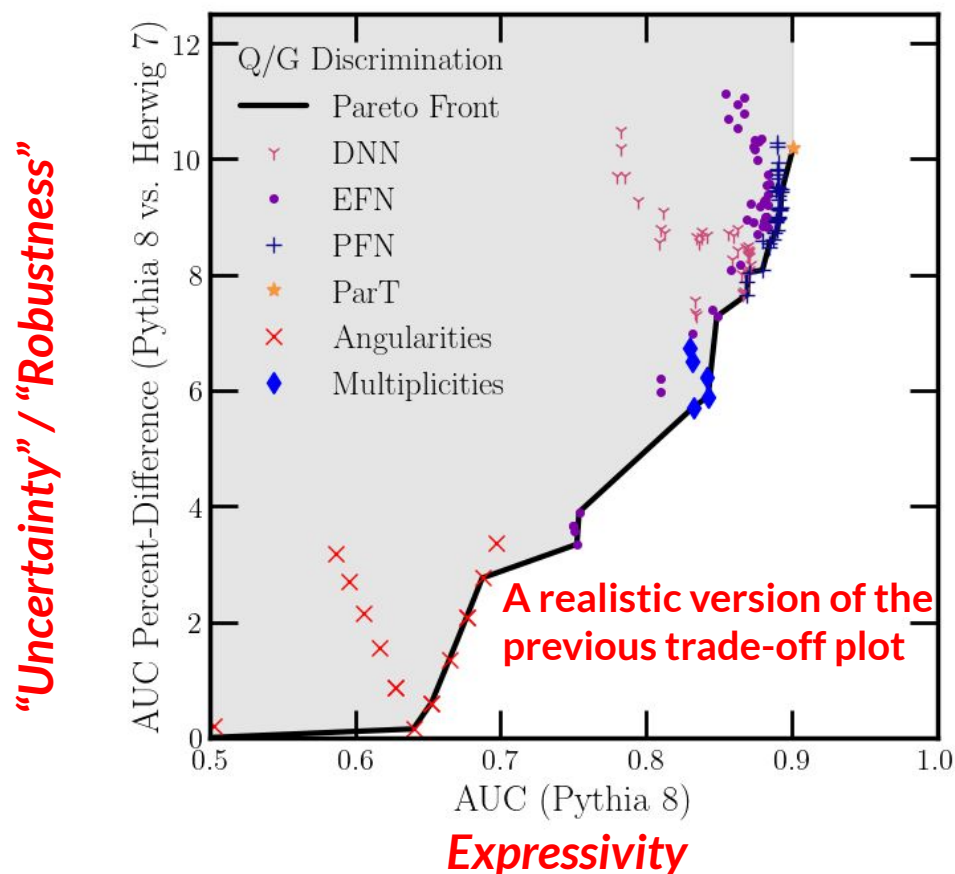
Interpretability is a choice.

Model *B* has built-in interpretability. Model *A* does not. That does not mean we should throw out Model *A*! **Interpretability vs. Expressivity is a trade off.**



Interpretability is a choice. Example

Interpretability vs. Expressivity is a trade off.



Less interpretable models (Transformers and large Deep Sets) get much higher performance on Pythia tagging, because they are more expressive.

More interpretable models (handcrafted features, EFNs) have worse performance because they are smaller and simpler.

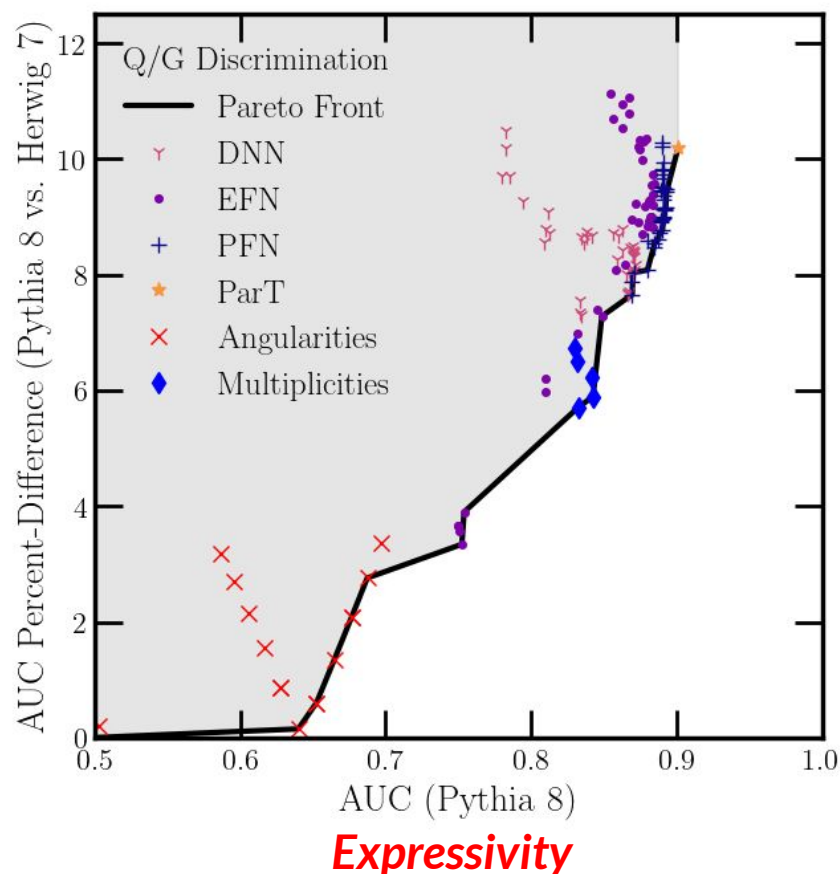
So why bother with the more interpretable model?

3. Do we even need interpretability?

Interpretability is a choice. Example

Interpretability vs. Expressivity is a trade off.

“Uncertainty” / “Robustness”



Because we don't just care about raw performance on Pythia. We care about performance on data.*

We want to guarantee that the tagger will act similarly on data and simulation, which requires some physics assumptions, which requires interpretability to check!

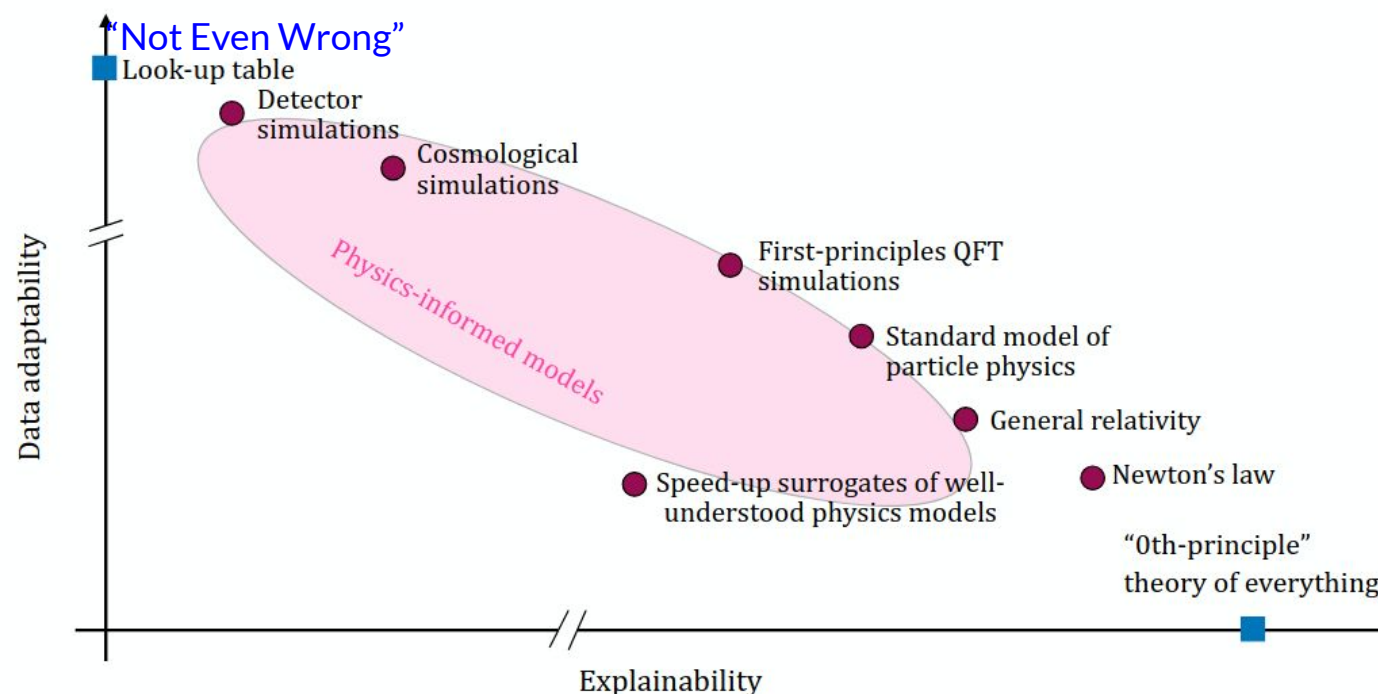
Because I know QCD, these handcrafted features / EFNs specifically target things I expect to be robust.

3. Do we even need interpretability?

*Here, Herwig is used as a proxy for real data.

Explainability is a choice.

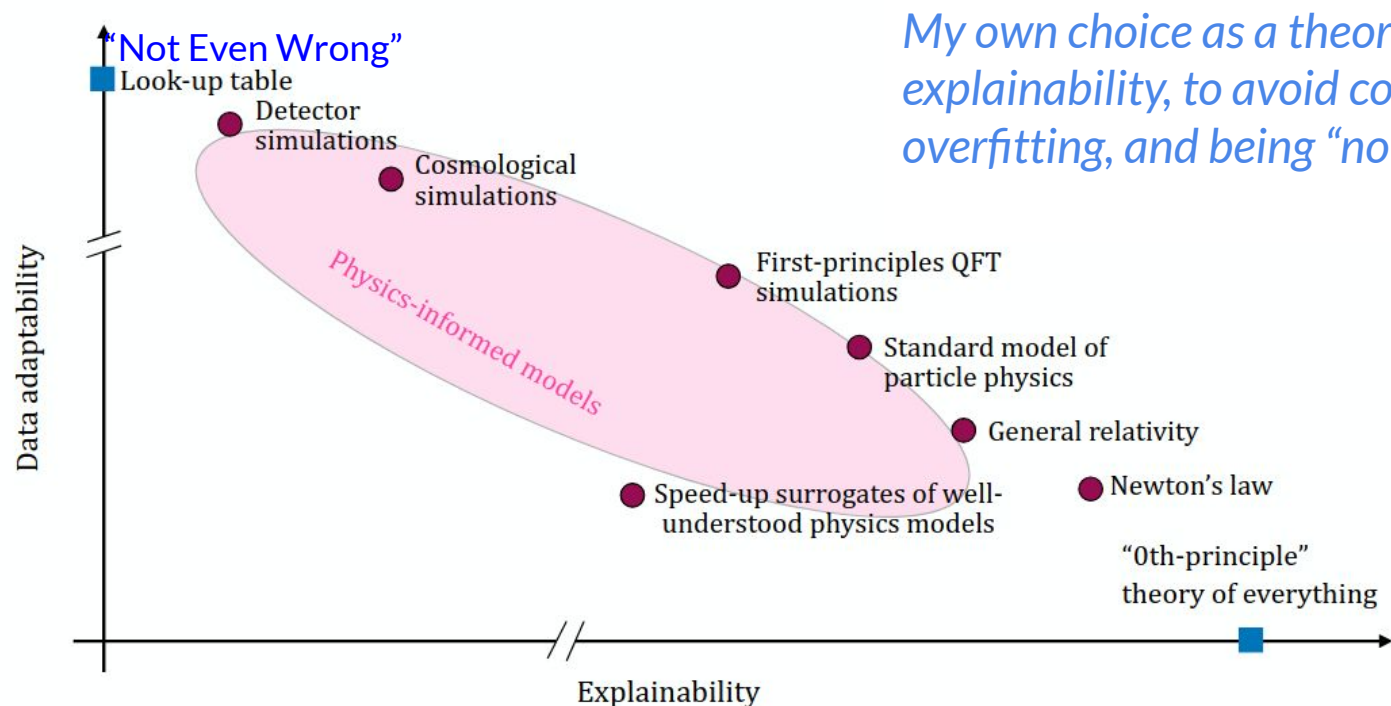
More **explainable** models are typically **less able to fit arbitrary data**. Whether that is good or bad is context dependent!



In order to be explainable, you have to risk your model potentially not being a good fit!

Explainability is a choice.

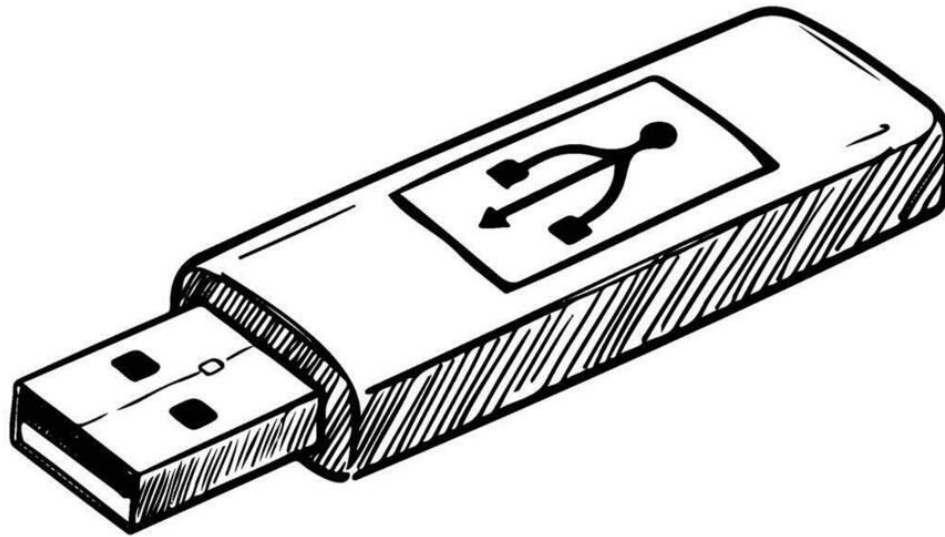
More **explainable** models are typically **less able to fit arbitrary data**. Whether that is good or bad is context dependent!



In order to be explainable, you have to risk your model potentially not being a good fit!

Do we really need interpretability?

Would you plug this into your computer and run whatever is on it?



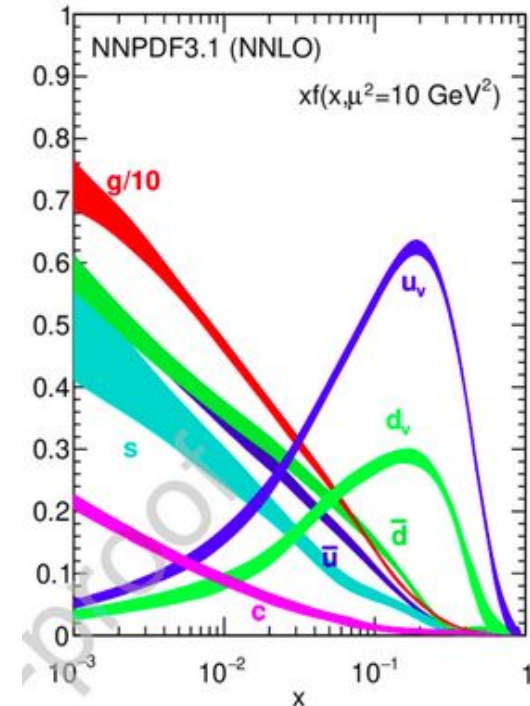
It has a 100% performant model on it!

Do we really need interpretability?

$$\sigma_{pp \rightarrow X} = \sum_{a,b} \int dx_1 dx_2 f_{a/p}(x_1, \mu_F) f_{b/p}(x_2, \mu_F) \hat{\sigma}_{a,b \rightarrow X}(\mu_F, \mu_R)$$

Detector simulations* are expensive. Anything we can do to make them faster is highly appreciated. Do we really care about how the internals of these surrogate models work? *Maybe a little, just so we can trust it works as experts expect?* See e.g. the CaloChallenge [[2410.21611](https://arxiv.org/abs/2410.21611)]

If you just need a flexible parametric fit to data as part of a model, you probably don't care about what the parametric form actually is. *Maybe a little, just so we trust the PDFs agree with theory expectations?* [e.g. [NN-PDFs](#)]



If you use an LLM as a surrogate search engine / debugger / therapist / paper writer, does it really matter how the internals work? *Maybe a little, just so we thrust the LLM won't do something absurd?*

My bias: I think we should avoid zero interpretability if possible. I don't want a model that I can't connect to anything! We can't parameterize away everything.

*Supposedly interpretable by experts, but not me!

Do we really need interpretability?

$$\sigma_{pp \rightarrow X} = \sum_{a,b} \int dx_1 dx_2 f_{a/p}(x_1, \mu_F) f_{b/p}(x_2, \mu_F) \hat{\sigma}_{a,b \rightarrow k}(\mu_F, \mu_R)$$

Detector simulations* are expensive. Anything we can do to make them faster is highly appreciated. Do we really care about how the internals of the works as experts expect

If you just need you probably Maybe a little, just so

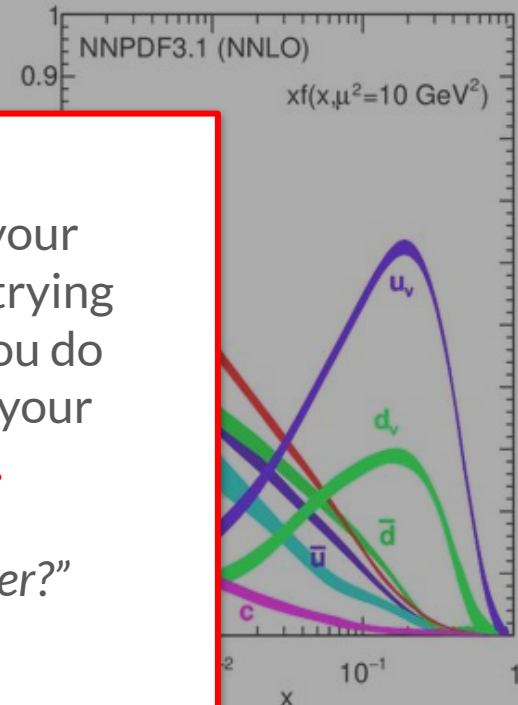
If you use an therapist / pa work? Maybe a li

Key Takeaway: Whether or not you need interpretability in the first place depends on your goals, and knowing precisely what it is you are trying to optimize for in the first place! Sometimes, you do want to optimize for understanding as part of your goal. **Task Specification Matters! So do it.**

“Do I want the most precise quark vs. gluon tagger?”

vs.

“Do I want a slightly suboptimal quark vs. gluon tagger I can understand?”



should avoid zero
don't want a model that
anything! We can't
parameterize away everything.

3. Do we even need interpretability?

*Supposedly interpretable by experts, but not me!

Conclusion

The main goal of this lecture was to **raise questions** that you should think about when applying ML and modeling yourself!

- What is my actual problem I am trying to optimize and why? Is it just a proxy for another downstream problem?
- Do I even need my model to be interpretable or explainable? How much? What guarantees can I make about my model's properties?
- What can and can't I interpret? What can and can't my audience interpret?
- What is the domain knowledge I am relating to?
- What would I do with the interpretation if I had it? What if it did or didn't agree with prior expectations?
- Is the information I try to explain using interpretability methods just confirmation bias?
- Can I build in structures I can interpret directly in my model?
- Will I use post-hoc methods? Which ones and when can they fail?

Please email me your questions and ideas at gambhirb@ucmail.uc.edu!

“The most merciful thing in the world, I think, is the inability of the human mind to correlate all its contents.”

- H.P. Lovecraft

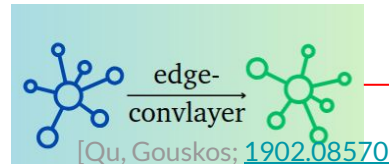
On attempting to interpret the most uninterpretable neural network of all.

Appendices

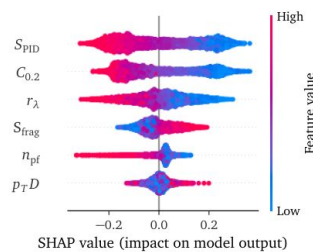
Post-Hoc Interpretability: Real Example

Given a large **Graph Neural Network** trained to distinguish top-quark jets from QCD jets, interpret it!

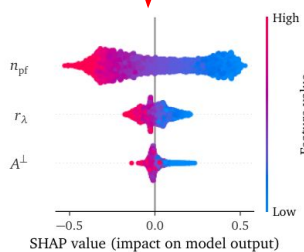
SHAP



PySR



n_{pf} strangely undervalued?
Against intuition!
Intervention: Look at decorrelated features



n_{pf} agrees with intuition!
The model roughly values the same things as us

I trust this model more now about what it says are the next few most important features.

$$\tanh^3 \left(0.55 \cdot C_{0.2} + 2 \cdot (-0.02 \cdot r_\lambda \cdot (C_{0.2} \cdot p_{TD} \cdot S_{PID} \cdot S_{frag} - 0.25) + 1) \right)^3$$

A post-hoc approximation of the graph neural network that is far more interpretable with only a minor performance hit!

observables	model	AUC	Rej _{30%}
$(n_{pf}, p_{TD}, C_{0.2}, r_\lambda, S_{PID}, S_{frag}, E_Q)$	MLP	0.872	66.87
	PySR	0.871	66.58

2. Why (and why not) interpretability?

Intrinsic Interpretability*

Models are a choice! One can simply choose to build interpretable structures into their model. Model A from earlier is an example of an intrinsically interpretable model.

Real Example: Deep Sets $\rightarrow$ Energy Flow Networks (EFNs)

Deep sets are arbitrary functions on point clouds.

$$F \left(\sum_{i=1}^M \Phi(p_i) \right)$$

EFNs are functions on point clouds with specific soft and collinear behavior.

$$F \left(\sum_{i=1}^M z_i \Phi(\hat{p}_i) \right)$$

*Bias Acknowledgement: My favorite type.

Intrinsic Interpretability*

Models are a choice! One can simply choose to build interpretable structures into their model. Model A from earlier is an example of an intrinsically interpretable model.

Real Example: Deep Sets → Energy Flow Networks (EFNs)

Deep sets are arbitrary functions on point clouds.

$$F \left(\sum_{i=1}^M \Phi(p_i) \right)$$

EFNs are functions on point clouds with specific soft and collinear behavior.

$$F \left(\sum_{i=1}^M z_i \Phi(\hat{p}_i) \right)$$

Even if I can't interpret the full function, I at least know how the EFN function must act in *some* limits! More interpretable!

*Bias Acknowledgement: My favorite type.

Intrinsic Interpretability*

Models are a choice! One can simply choose to build interpretable structures into their model. Model A from earlier is an example of an intrinsically interpretable model.

Real Example: Deep Sets → Energy Flow Networks (EFNs)

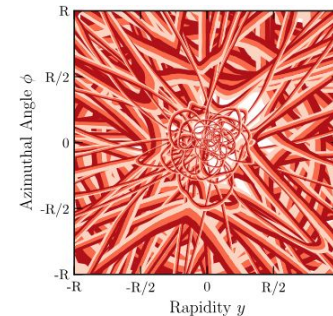
Deep sets are arbitrary functions on point clouds.

(Cannot draw a picture!)

$$F \left(\sum_{i=1}^M \Phi(p_i) \right)$$

Because the EFN is 2D *by construction*, we can draw pictures!

EFNs are functions on point clouds with specific soft and collinear behavior.



Latent space contours

$$F \left(\sum_{i=1}^M z_i \Phi(\hat{p}_i) \right)$$

*Bias Acknowledgement: My favorite type.

Interpretability is a choice.

Model *B* has built-in interpretability. Model *A* does not. That does not mean we should throw out Model *A*! **Interpretability vs. Expressivity is a trade off.**

Model *A* is *more expressive*. If all we really care about is predicting the motions of planets accurately, then Model *A* will simply predict them more accurately.

Perhaps the true theory of gravity is some terribly complex thing that only a network can fit. Who are we to try to interpret it?

You choose. Do you want something highly accurate, or a simpler model you can trust?

Key Takeaway: Interpretability is a modeling decision. When designing a model, *you* choose how interpretable to make it, at the cost of expressivity. You also get to decide what *you* can interpret. This is no different than designing the functional form of any classical fit.

My own choice as a theorist is to err on the side of models with built-in interpretability, so we can guarantee and verify the model will behave exactly as intended, as we will see...

3. Do we even need interpretability?

Interpretability and Problem Specification

The choice of model, and therefore the way it is interpreted, depends heavily on the problem we are trying to solve. *Often, the problem we are currently solving isn't actually the problem we deeply care about.*

"I want a good quark/gluon tagger"

"What exactly do you mean, `good`?"

"Let's say a high AUC on a PYTHIA test set"

"Here is a perfect loop-up table for all of PYTHIA. It has a perfect AUC of 1.00!"

"No! That's not really what I wanted. I also want it to do well on data."

"Then you need to build a model that will capture the features of PYTHIA you also expect in data. PYTHIA AUC isn't really what you want — why do you even need this tagger?"

"I need to tag quarks and gluons so I can measure QCD parameters with better precision."

"Then don't optimize for PYTHIA AUC. Optimize for your final downstream precision!"

3. Do we even need interpretability?

Optimizing for the *Real* Downstream Task

Suppose I care about measuring a quark-gluon mixing fraction parameter. Then I shouldn't optimize for tagging performance — that's just a surrogate. *Just optimize directly for the thing I care about!*

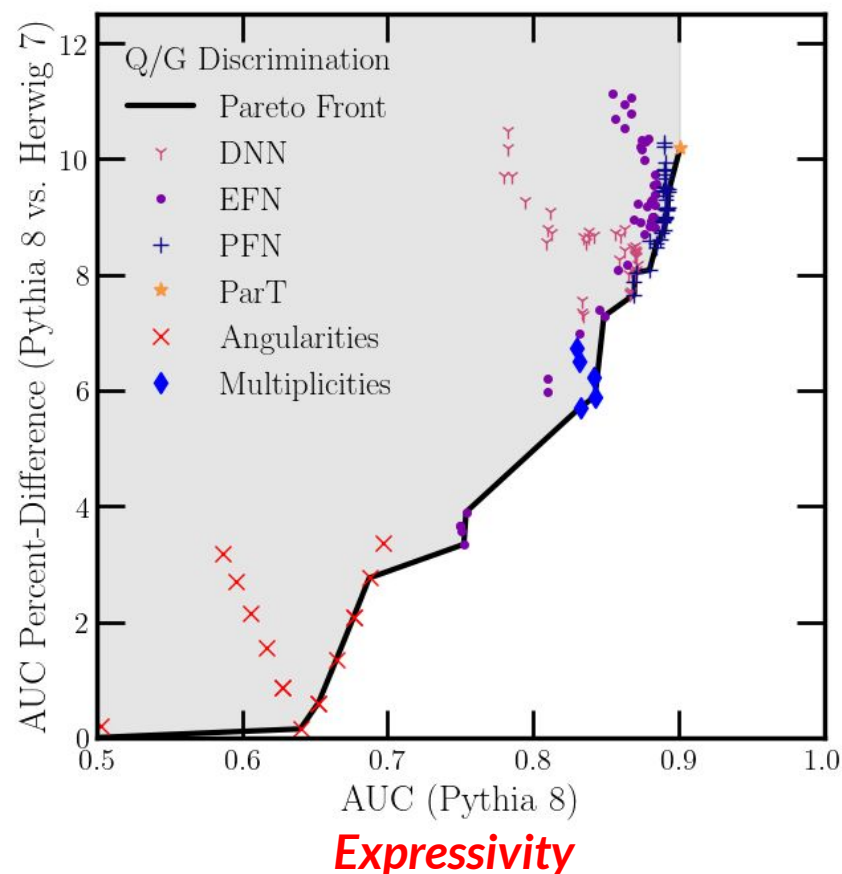
Classifier	True κ	Pythia 8	Herwig 7		Result
		Inferred $\mathbb{E}[\hat{\kappa}]$	Inferred $\mathbb{E}[\hat{\kappa}]$	Calibrated $\mathbb{E}[\hat{\kappa}]$	
Large PFN	0.50	0.490 ± 0.005	0.296 ± 0.007	0.529 ± 0.006	Biased ✗
	0.25	0.253 ± 0.005	0.125 ± 0.005	0.305 ± 0.006	Biased ✗
Small PFN	0.50	0.504 ± 0.013	0.336 ± 0.016	0.478 ± 0.017	Unbiased ✓
	0.25	0.258 ± 0.013	0.157 ± 0.014	0.268 ± 0.013	Unbiased ✓

Smaller models, which are naively worse taggers, actually lead to *less* bias on the final inferred parameter after calibration to 'data'!

Interpretability is a choice. **Example**

Interpretability vs. Expressivity is a trade off.

“Uncertainty” / “Robustness”



My favorite ML problem: Given a jet (a spray of particles) at the LHC, what particle initiated the jet?

We can define models f that take in jet features x (either particle level or hand-crafted features) and return the jet flavor.

These models are often trained on simulation (Pythia) and evaluated on collider data.

3. Do we even need interpretability?